Indian Institute of Technology, Bhilai

CSP203: Software Tools & Technologies Laboratory

Lab Activity - 5

Full Marks: 35 *Deadline: 11:59 PM, 19-08-2025*

Submission Instructions:

1) Answer all the questions.
2) Write all the scripts, execute and take screenshots of the scripts, name the scripts files as <Your_Roll_No>_<Lab_Activity_No>_<Question_No>.
3) Prepare a report pdf which contains the question number and screenshot of script code and corresponding outputs. Report quality will contain **5 marks**. You should prepare a zip folder which will contain the report and all the script files.
4) Naming of the zip file should strictly follow the given format:
   <Roll_No>_<Name>_<Activity_No>
   If your name is Shareen, roll number is B24CS055 then the filename should be:
   B24CS055_Shareen_04.pdf
5) **Plagiarism is strictly prohibited.**

Objectives:

1) Understand how to define and use functions in Bash scripts to modularize code.
2) Apply array handling and parameter passing in Bash to find maximum values.
3) Practice using Linux commands for processing file data.
4) Learn how to use regular expressions and string processing tools for input validation.
5) Enhance problem-solving and scripting skills by validating real-world inputs like URL/link.

Task 1. Write a Bash script that uses functions to find the maximum element in an array.
Requirements:

1. The script must define two functions:
2. A main function that:
    a. Prompts the user for the number of elements.
    b. Reads the elements of the array from the user.
    c. Prints the maximum element
3. A greatest function that:
    a. Takes the array as input (via arguments).
    b. Calculates and returns the maximum element to the main

The main function should call the greatest function with the array as an argument. [Marks 10]

Task 2. Write a bash script to take file name as input. If the filename exists then print the number of lines, number of words, number of characters of the file otherwise print "404: file not found". [Marks 10]

Task 3. Write a Bash script to validate whether a given input is a valid URL/link.The script should:

1. Prompt the user to enter a URL.
2. Use a regular expression to check that the input follows the correct URL format, such as:
    a. Starts with http:// or https://
    b. Followed by a valid domain name
3. If the format is valid, use the curl or wget command to check whether the link is reachable.
4. Display:
    a. "Valid and reachable link." if all checks pass
    b. "Invalid link format." if the input does not match the regex
    c. "Link format valid but unreachable." if the link exists but cannot be reached

[Marks 10]